A FUNCTORIAL ACYCLIC REPLACEMENT KILLING A PRESCRIBED SET OF COMPLEXES

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file sites-cohomology.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The proof is the author's complete construction, each of the two assertions being verified in the span; the closing sentence records that the second assertion holds by construction.

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Editorial note. The preamble is the chapter's own retained minimal preamble (sources/stacks-sites-cohomology/retained-preamble.tex), a verbatim excerpt of the native preamble: the Xy-pic setup and the native theorem declarations with the mathematical macros.

Allowed standard prerequisites (editorial).

- Functorial injective embeddings. Same chapter, Theorem theorem-injective-embedding-grothendieck. Supplies the functorial embeddings of the terms into injective objects out of which the replacement is built.
- Products of injective objects and their mapping properties. Homology, the product rule for injective objects and the description of the relevant two-term complexes. Supply that products of the chosen injective objects are injective and that the maps out of the prescribed acyclic complexes into them vanish on homotopy, which is what makes the prescribed complexes redundant.

Lemma 0.1. *Let $\mathcal{A}$ be a Grothendieck abelian category. Let $(K_i^\bullet)_{i\in I}$ be a set of acyclic complexes. There exists a functor $M^\bullet \mapsto \mathbf{M}^\bullet(M^\bullet)$ and a natural transformation $j_{M^\bullet} : M^\bullet \to \mathbf{M}^\bullet(M^\bullet)$ such*

(1) *$j_{M^\bullet}$ is a (termwise) injective quasi-isomorphism, and*
(2) *for every $i \in I$ and $w : K_i^\bullet \to M^\bullet$ the morphism $j_{M^\bullet} \circ w$ is homotopic to zero.*

Proof. For every $i \in I$ choose a (termwise) injective map of complexes $K_i^\bullet \to L_i^\bullet$ which is homotopic to zero with $L_i^\bullet$ quasi-isomorphic to zero. For example, take $L_i^\bullet$

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to be the cone on the identity of $K_i^\bullet$. We define $\mathbf{M}^\bullet(M^\bullet)$ by the following pushout diagram

$$\begin{array}{ccc} \bigoplus_{i\in I}\bigoplus_{w:K_i^\bullet\to M^\bullet} K_i^\bullet & \longrightarrow & M^\bullet \\ \downarrow & & \downarrow \\ \bigoplus_{i\in I}\bigoplus_{w:K_i^\bullet\to M^\bullet} L_i^\bullet & \longrightarrow & \mathbf{M}^\bullet(M^\bullet). \end{array}$$

Then $M^\bullet \mapsto \mathbf{M}^\bullet(M^\bullet)$ is a functor. The right vertical arrow defines the functorial injective map $j_{M^\bullet}$. The cokernel of $j_{M^\bullet}$ is isomorphic to the direct sum of the cokernels of the maps $K_i^\bullet \to L_i^\bullet$ hence acyclic. Thus $j_{M^\bullet}$ is a quasi-isomorphism. Part (2) holds by construction. $\square$